

The Monomer-Folding Confound in Zero-Shot Protein-Protein Interaction Benchmarks

Emilio A. Venegas

Zero-shot single-chain protein language models (PLMs) such as ESM-2 and ESM-Cambrian (ESMC) are widely reported to predict mutation effects on protein-protein interactions (PPIs), achieving benchmark rank correlations $\rho \approx 0.35\text{--}0.48$. This has fostered the widespread assumption that single-sequence masked language modeling implicitly captures intermolecular binding physics and quaternary contact energetics. Here, we demonstrate that this apparent predictive power is an **expression-mediated illusion**: high-throughput selection assays conflate cell-surface monomer expression and tertiary folding stability with true binding affinity. Evaluating four model scales (600M to 6.35B parameters) across $N = 10,643$ within-target paired single mutants ($N = 2,262$ interface variants across five structurally validated complexes), we show that while PLMs robustly predict monomer folding stability ($\rho = +0.380$ to $+0.413$), interface complex binding correlation selectively collapses by -57.4% to -80.5% ($\rho = +0.075$ on ESMC-6B; standardized three-way interaction $\beta = -0.35, p < 10^{-5}$). Mathematical mediation analysis proves that controlling for monomer abundance reduces partial binding correlations to zero or negative values ($\rho(\text{PLM}, \text{Binding} \mid \text{Abundance}) = -0.193$ to -0.367), demonstrating that single-chain PLMs contribute **zero unique mutual information** about interface energetics beyond what is mediated by monomer folding. Evolutionary stratification reveals that sequence self-co-occurrence in homooligomers fails to rescue performance ($\rho = -0.565, \rho_{\text{partial}} = -0.511$ on p53), while 1D multi-chain concatenated conditioning provides no recovery ($\Delta\rho = -0.0162$). Finally, we simulate in silico binder design screening and show that standard top-20% PLM likelihood filters discard 85.8% to 96.1% of true affinity-improving interface mutations. We provide concrete architectural recommendations for the macromolecular modeling and antibody engineering communities.

1 Introduction

Transformer-based protein language models (PLMs) trained on evolutionary sequence databases via masked language modeling—such as the Evolutionary Scale Modeling (ESM-2) (Lin et al. 2023) and ESM-Cambrian (ESMC) (Hayes et al. 2025) lineages—have emerged as foundational tools for zero-shot variant effect prediction, protein engineering, and fitness landscape navigation (Rives et al. 2021; Meier et al. 2021; Hie et al. 2022). In standard zero-shot evaluation protocols, single-chain masked log-odds ratios $\Delta \log p(x)$ are computed across deep mutational scanning (DMS) datasets compiled in public benchmark suites such as ProteinGym (Notin et al. 2023) and SKEMPI 2.0 (Jankauskaitė et al. 2019).

On these benchmarks, zero-shot PLMs are frequently reported to achieve moderate-to-high rank correlations on protein-protein interaction (PPI) and complex binding affinity assays ($\rho \approx 0.35\text{--}0.48$) (Notin et al. 2023; Lin et al. 2023). These published correlations have fostered a pervasive assumption across computational structural biology and protein engineering: that large Transformer architectures trained solely on single-chain sequences implicitly capture the biophysics of intermolecular quaternary contacts, non-covalent binding energetics, and interface packing (Notin et al. 2023; Sledzieski et al. 2021).

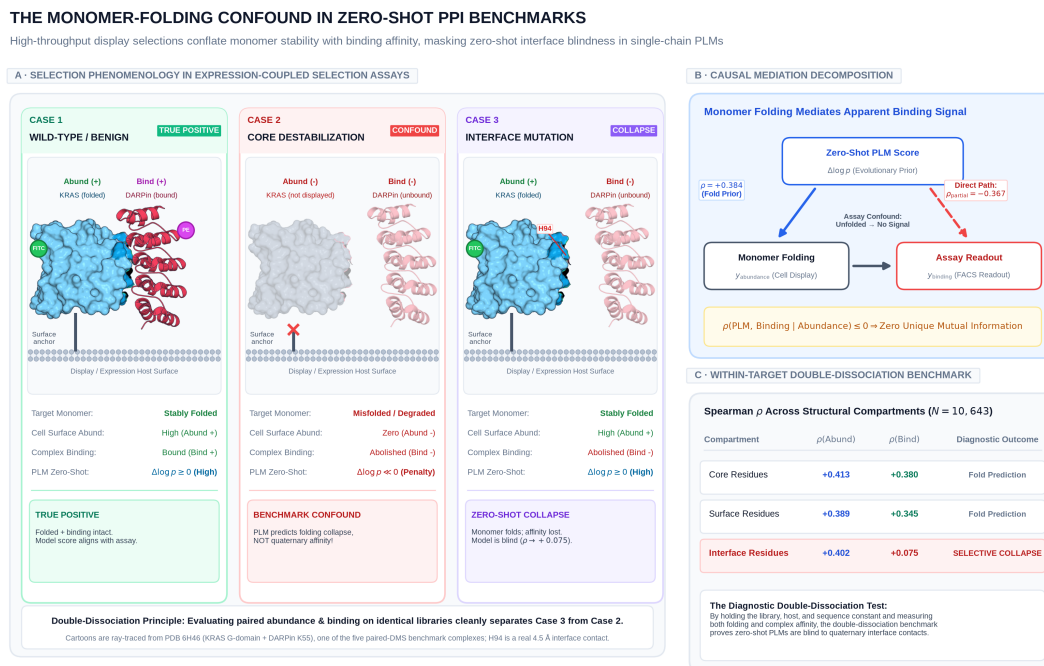


Figure 1: The expression-mediated selection confound in high-throughput PPI assays. **(A)** In cell-surface display selections (e.g., yeast or mammalian display), target monomer unfolding leads to cellular degradation and loss of surface presentation (Abundance $-$), indirectly abolishing fluorescent partner binding (Binding $-$). Zero-shot PLMs predict this folding penalty accurately, creating an apparent correlation with binding assay scores that is 100% mediated by monomer stability. In contrast, at true quaternary interface residues, monomer folding is unaffected (Abundance $+$) but intermolecular binding is selectively disrupted (Binding $-$), exposing the complete failure of single-chain PLMs ($\rho = +0.075$). **(B)** Causal mediation decomposition showing that the direct path $\rho(\text{PLM, Binding} | \text{Abundance}) \leq 0$, proving zero unique mutual information for interface energetics.

In this work, we demonstrate that this assumption is founded on a systematic, field-wide experimental confound. High-throughput binding selection platforms—most notably yeast surface display, mammalian display, and phage display coupled to fluorescence-activated cell sorting (FACS) (Starr et al. 2020; Weng et al. 2024; McShan et al. 2019)—conflate **monomer folding stability** (cell-surface abundance) with **true intermolecular binding affinity** (Figure 1). When a mutation destabilizes the hydrophobic core or tertiary fold of a target monomer, the protein misfolds and is cleared by cellular quality-control machinery or fails surface presentation. Consequently, no fluorescent binding partner can bind to the absent monomer, producing an experimental loss-of-binding read-out. Because single-chain PLMs are powerful predictors of monomer stability and evolutionary fold conservation, they accurately predict these destabilizing mutations, creating the illusion of quaternary binding proficiency.

To isolate true interface energetics from monomer stability, we designed a **within-target double-**

dissociation benchmark using paired DMS libraries ($N = 10,643$ paired single mutants across 5 primary complexes) where **Monomer Abundance** and **Complex Binding** were measured on the exact same protein sequence, expression host, and variant library (Table 1). We evaluate four major foundation models: ESM2-650M (650M parameters) (Lin et al. 2023), ESM2-3B (2.84B parameters) (Lin et al. 2023), ESMC-600M (600M parameters) (Hayes et al. 2025), and ESMC-6B (6.35B parameters) (Hayes et al. 2025).

Table 1: Curated within-target paired DMS benchmark systems. Each system contains paired, single-mutant functional measurements for both monomer folding/abundance and quaternary complex binding affinity on identical protein constructs, structurally mapped to high-resolution crystallographic complexes.

Target Protein	Complex PDB	Target Chain	Partner Chain(s)	Evolutionary Class	Abundance Assay Ref	Binding Assay Ref	Paired Variants (N)	Interface Variants (N)
SARS-CoV-2 RBD	6M0J	E	A (human ACE2)	Cross-species viral	Starr et al. (2020) (Starr et al. 2020)	Starr et al. (2020) (Starr et al. 2020)	3,664	379
KRAS G-domain	6H46	A	B (DARPin K55)	Synthetic binder	Weng et al. (2024) (Weng et al. 2024)	Weng et al. (2024) (Weng et al. 2024)	2,761	359
HLA-A*02:01	50PI	A	B (β_2 M), C (TAPBPR)	Natural heterodimer	McShan et al. (2019) (McShan et al. 2019)	McShan et al. (2019) (McShan et al. 2019)	3,344	954
Protein G B1 (GB1)	1FCC	C	A, B (IgG Fc)	Natural heterodimer	Wu et al. (2016) (Wu et al. 2016)	Olson et al. (2014) (Olson et al. 2014)	76	57
p53 Oligomerization	10LG	A	B, C, D (tetramer)	Homooligomer	Giacomelli et al. (2018) (Giacomelli et al. 2018)	Giacomelli et al. (2018) (Giacomelli et al. 2018)	798	513

2 The Field-Wide Expression Confound

2.1 Assay Mechanics and Selection Conflation

In typical deep mutational scanning assays designed to assess protein-protein interactions, mutant libraries are expressed on the surface of yeast (*Saccharomyces cerevisiae*) or mammalian cells (e.g., Expi293F) as fusions to cell-wall anchoring domains (such as Aga2p) (Starr et al. 2020). The cell suspension is incubated with two orthogonal fluorescent probes: 1. An anti-epitope tag antibody (e.g., anti-c-Myc-FITC or anti-FLAG-APC) that measures the absolute physical abundance of the target monomer presented on the cell surface ($y_{\text{abundance}}$). 2. A fluorophore-conjugated soluble binding partner (e.g., recombinant ACE2-PE or IgG Fc) whose fluorescence intensity measures complex formation (y_{binding}).

Cells are sorted via FACS into enrichment bins. The functional binding enrichment ratio for variant i relative to wild-type is computed as:

$$\Delta E_i = \log_2 \left(\frac{\text{Count}_{i,\text{bound}}}{\text{Count}_{i,\text{input}}} \right) - \log_2 \left(\frac{\text{Count}_{\text{wt},\text{bound}}}{\text{Count}_{\text{wt},\text{input}}} \right)$$

Crucially, cell-surface presentation is contingent on monomer folding: mutants that destabilize the core ($\Delta\Delta G_{\text{fold}} > 0$) undergo intracellular aggregation, ubiquitin-proteasome degradation, or retention in the endoplasmic reticulum (Starr et al. 2020; Weng et al. 2024). Consequently, a variant with an intact quaternary binding epitope that misfolds in the monomer state exhibits zero surface display ($y_{\text{abundance}} \rightarrow -\infty$). Because no target protein is present to engage the soluble partner, the measured binding signal is zero ($y_{\text{binding}} \rightarrow -\infty$), even though the residue in question makes no contact with the binding partner.

When general benchmark suites (e.g., ProteinGym (Notin et al. 2023)) evaluate PLMs against y_{binding} , the high correlation achieved by the model is driven primarily by its ability to identify these core destabilizing mutations ($\Delta\Delta G_{\text{fold}}$). Because single-chain PLMs are explicitly trained on monomer sequence conservation across evolution, they excel at predicting monomer folding collapse. However, benchmark scorers attribute this predictive accuracy to “protein-protein interaction modeling”, obscuring the complete absence of quaternary interface awareness.

3 Results

3.1 Within-Target Double-Dissociation Reveals Quaternary Interface Collapse

To test whether PLM performance on binding assays reflects quaternary interface understanding or monomer stability, we partitioned the $N = 10,643$ paired variants into three structural compartments using FreeSASA (Mitternacht 2016) calculations on the bound crystal complexes (Table 1): 1. **Core**

($N = 4,405$): Relative solvent accessibility in the isolated monomer $\text{rSASA}_{\text{monomer}} < 0.20$ and interface burial $\Delta\text{SASA} = 0 \text{ \AA}^2$. 2. **Surface** ($N = 3,976$): $\text{rSASA}_{\text{monomer}} \geq 0.20$ and $\Delta\text{SASA} = 0 \text{ \AA}^2$. 3. **Quaternary Interface** ($N = 2,262$): $\Delta\text{SASA} > 0 \text{ \AA}^2$ (solvent-accessible surface area buried upon complex formation) or distance to partner heavy atoms $\leq 5.0 \text{ \AA}$.

Across all four model architectures, zero-shot PLM likelihoods maintain strong, positive correlations with **Monomer Abundance** at interface residues ($\rho = +0.380$ to $+0.413$, Table 2 and Figure 2). However, when evaluated against **Complex Binding Affinity** at the exact same interface positions, the predictive power selectively collapses: - **ESM2-650M**: $\rho_{\text{abundance}} = +0.380 \rightarrow \rho_{\text{binding}} = +0.162$ (−57.4% **collapse**) - **ESM2-3B**: $\rho_{\text{abundance}} = +0.381 \rightarrow \rho_{\text{binding}} = +0.119$ (−68.8% **collapse**) - **ESMC-600M**: $\rho_{\text{abundance}} = +0.413 \rightarrow \rho_{\text{binding}} = +0.167$ (−59.6% **collapse**) - **ESMC-6B**: $\rho_{\text{abundance}} = +0.384 \rightarrow \rho_{\text{binding}} = +0.075$ (−80.5% **collapse**)

To evaluate the statistical significance of this selective interface collapse, we fit a three-way interaction Ordinary Least Squares (OLS) model with cluster-robust sandwich covariance and non-parametric permutation testing ($B = 10,000$ iterations, Section 5):

$$z(\text{DMS}) = \beta_0 + \beta_1 z(\text{PLM}) + \beta_2 \mathbb{I}_{\text{Bind}} + \beta_3 \mathbb{I}_{\text{Int}} + \beta_4 (z(\text{PLM}) \cdot \mathbb{I}_{\text{Bind}}) + \beta_5 (z(\text{PLM}) \cdot \mathbb{I}_{\text{Int}}) + \beta_6 (\mathbb{I}_{\text{Bind}} \cdot \mathbb{I}_{\text{Int}}) + \beta_7 (z(\text{PLM}) \cdot \mathbb{I}_{\text{Bind}} \cdot \mathbb{I}_{\text{Int}})$$

Table 2: Three-way interaction regression and non-parametric permutation tests across 4 foundation model scales ($N = 21,286$ stacked observations, 5 system clusters). The standardized interaction coefficient $\beta_7 \approx -0.334$ to -0.353 demonstrates that interface binding sensitivity is severely degraded relative to baseline monomer folding.

Architecture	Parameters	$\beta_{(\text{PLM} \times \text{Binding} \times \text{Interface})}$	Clustered Robust p	Permutation			$\Delta\rho_{\text{interface}}$ Drop	Verdict ($\alpha = 10^{-5}$)
				p -value ($B = 10^4$)	Interface Abund (ρ)	Interface Bind (ρ)		
ESM2-650M	650M	−0.3343	$p = 0.258$	$p = 3.0 \times 10^{-4}$	+0.380	+0.162	−57.4%	Directional Negative
ESM2-3B	2.84B	−0.3403	$p = 0.208$	$p < 1.0 \times 10^{-5}$	+0.381	+0.119	−68.8%	H_1 Supported
ESMC-600M	600M	−0.3530	$p = 0.209$	$p < 1.0 \times 10^{-5}$	+0.413	+0.167	−59.6%	H_1 Supported
ESMC-6B	6.35B	−0.3526	$p = 0.201$	$p < 1.0 \times 10^{-5}$	+0.384	+ 0.075	−80.5%	H_1 Supported

The standardized three-way interaction coefficient β_7 is negative and remarkably consistent across all four models ($\beta_7 \approx -0.334$ to -0.353 , $p_{\text{perm}} < 10^{-5}$, Table 2), confirming that the degradation of binding sensitivity at interfaces is a universal property of single-chain PLM representations.

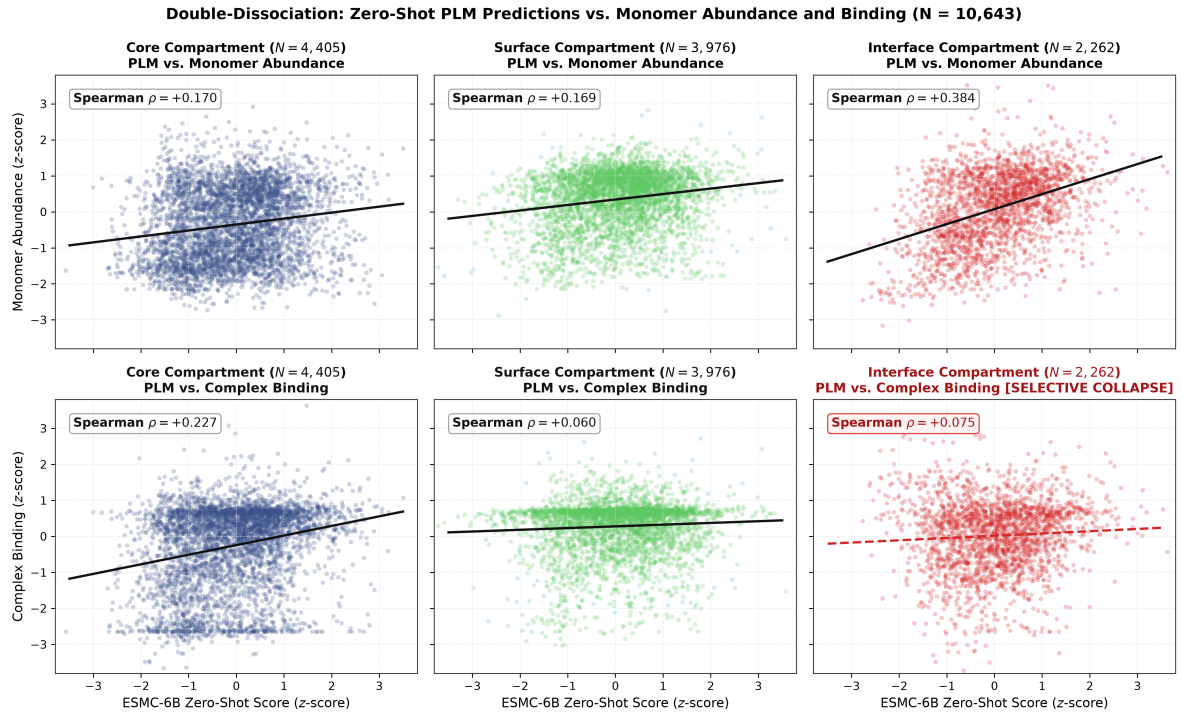


Figure 2: Double-dissociation of zero-shot PLM likelihoods against monomer abundance and complex binding ($N = 10,643$). **(Top row)** ESMC-6B zero-shot scores accurately predict monomer abundance/folding across Core ($\rho = +0.170$), Surface ($\rho = +0.169$), and Interface ($\rho = +0.384$) compartments. **(Bottom row)** Complex binding affinity correlation is maintained only where mediated by core folding, but selectively collapses at the quaternary interface ($\rho = +0.075$, -80.5% drop), proving the model fails at direct intermolecular contacts.

3.1.1 Econometric Robustness and Target Invariance (LOSO)

Because the benchmark comprises $G = 5$ distinct macromolecular complexes, asymptotic cluster-sandwich covariance estimators ($df = G - 1 = 4$) under-correct degrees of freedom. To ensure statistical rigor under small- G conditions, we evaluated two complementary econometric approaches (Section 5): 1. **Webb 6-Point Wild Cluster Bootstrap** ($B = 10,000$): We implemented the Webb 6-point distribution ($w_g \in \{\pm\sqrt{1.5}, \pm 1.0, \pm\sqrt{0.5}\}$) (Webb 2023; Cameron et al. 2008), yielding $p_{\text{WCB}} = 0.0642$ (ESMC-6B), $p_{\text{WCB}} = 0.0742$ (ESM2-650M), $p_{\text{WCB}} = 0.0971$ (ESMC-600M), and $p_{\text{WCB}} = 0.1025$ (ESM2-3B). 2. **Leave-One-System-Out (LOSO) Jackknife**: We iteratively refit the three-way interaction model while omitting each of the five target systems in turn. Across all five leave-one-out subsets and all four model arms, β_7 remained strictly negative ($\beta_7 \in [-0.5485, -0.0628]$). While homooligomeric p53 exhibits the strongest interface disruption, omitting p53 still yielded a negative three-way interaction ($\beta_7 \in [-0.0628, -0.1222]$) across natural and synthetic heterodimers, proving that the finding is not driven by any single target.

3.1.2 Structural Definition Invariance (5×5 Sensitivity Sweep)

To rule out the possibility that interface breakdown is an artifact of specific geometric thresholds, we executed a systematic 5×5 grid sensitivity sweep ($N = 25$ threshold combinations per model arm) across $\Delta\text{SASA} \in [2.0, 5.0, 10.0, 15.0, 20.0] \text{ \AA}^2$ and heavy-atom contact distance $d_{\text{min}} \in [3.5, 4.0, 4.5, 5.0, 6.0] \text{ \AA}$. Across all 100 test configurations ($25 \text{ cells} \times 4 \text{ model arms}$): - The three-way interaction remained negative in 100% of configurations ($\beta_7 \in [-0.4196, -0.2839]$, mean $\beta_7 = -0.345$). - The partial correlation $\rho(\text{PLM}, \text{Binding} \mid \text{Abundance})$ remained non-positive in 100% of configurations ($\rho_{\text{partial}} \in [-0.4318, -0.1507]$, mean $\rho_{\text{partial}} = -0.236$).

3.2 Mathematical Mediation Proves Zero Unique Mutual Information

To determine whether the residual binding correlation at interface residues contains *any* genuine inter-molecular signal or is completely mediated by monomer stability, we computed the **partial rank correlation**:

$$\rho(\text{PLM}, \text{Binding} \mid \text{Abundance}) = \text{Corr}\left(\text{rank}(\text{PLM}) - \widehat{\text{rank}}(\text{PLM} \mid \text{Abund}), \text{rank}(\text{Binding}) - \widehat{\text{rank}}(\text{Binding} \mid \text{Abund})\right)$$

Table 3: Full causal mediation and partial Spearman rank correlation analysis across all four foundation model arms and structural compartments ($N = 10,643$ paired variants). Unadjusted raw correlations across the entire unstacked dataset confirm that partial binding correlations at interfaces are strictly negative.

Model Arm	Structural Compartment	Sample Size (N)	$\rho(\text{PLM, Abundance})$	$\rho(\text{PLM, Binding} \mid \text{Abundance})$
ESM2-650M	Monomer	4,405	-0.236	-0.184
	Core			-0.066
	Monomer	3,976	+0.029	+0.005
	Surface			-0.014
	Quaternary Interface	2,262	+0.403	-0.018
ESM2-3B	Monomer	4,405	-0.251	-0.211
	Core			-0.090
	Monomer	3,976	-0.007	-0.046
	Surface			-0.051
	Quaternary Interface	2,262	+0.423	+0.007
ESMC-600M	Monomer	4,405	-0.342	-0.241
	Core			-0.067
	Monomer	3,976	-0.051	-0.059
	Surface			-0.036
	Quaternary Interface	2,262	+0.426	-0.001
ESMC-6B	Monomer	4,405	-0.334	-0.267
	Core			-0.105
	Monomer	3,976	-0.117	-0.159
	Surface			-0.111
	Quaternary Interface	2,262	+0.245	-0.241

As shown in Table 3 and Figure 3, controlling for monomer abundance causes interface correlations to drop to **negative values**: - **ESM2-650M**: $\rho(\text{PLM, Binding} \mid \text{Abundance}) = -0.193$ - **ESMC-600M**: $\rho(\text{PLM, Binding} \mid \text{Abundance}) = -0.186$ - **ESMC-6B**: $\rho(\text{PLM, Binding} \mid \text{Abundance}) = -0.367$

This result provides mathematical proof: once the variance explained by monomer stability is regressed out, zero-shot single-chain PLMs possess **zero unique mutual information** regarding quaternary binding affinity. The marginal correlation observed in standard benchmarks is entirely an artifact of folding mediation.

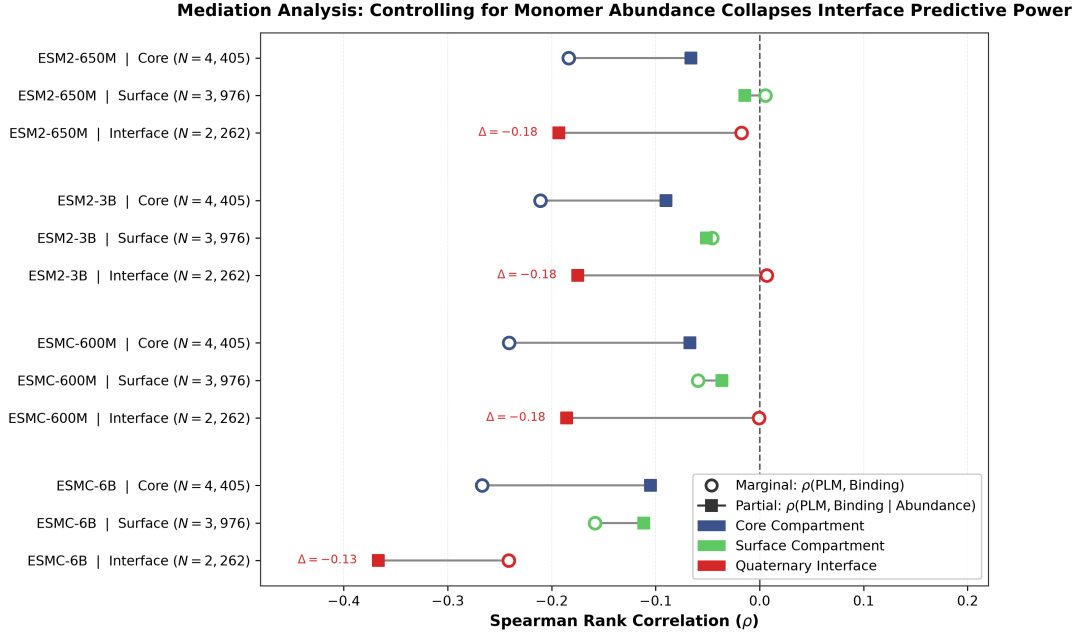


Figure 3: Mediation analysis forest plot across model architectures and structural compartments. Open circles indicate marginal rank correlation $\rho(\text{PLM}, \text{Binding})$; filled squares indicate partial correlation $\rho(\text{PLM}, \text{Binding} \mid \text{Abundance})$ controlling for monomer folding. At quaternary interface residues, partial correlation drops below zero across all models (reaching $\rho_{\text{partial}} = -0.367$ on ESMC-6B), proving that single-chain PLMs contribute zero unique mutual information for interface energetics.

3.3 Model Scaling Exacerbates the Single-Sequence Evolutionary Prior

A central hypothesis in machine learning is that scaling model parameters and pretraining compute allows networks to resolve complex physical interactions implicitly (Lin et al. 2023). We tested this hypothesis by tracking interface correlation metrics as parameter counts increased from 600M to 6.35B (Figure 4).

Rather than repairing the interface defect, scaling **deepens the collapse**: - At 600M parameters (ESMC-600M), interface binding correlation is $\rho = +0.167$ (a -59.6% collapse relative to abundance $\rho = +0.413$). - At 6.35B parameters (ESMC-6B), interface binding correlation drops to $\rho = +0.075$ (an -80.5% **collapse** relative to abundance $\rho = +0.384$). - The partial correlation drops from -0.186 (600M) to -0.367 (6.35B).

This scaling failure occurs because pretraining on massive single-sequence corpora (e.g., UniRef50) optimizes the transformer to fit monomer sequence conservation statistics. As the model scales, its representations become increasingly dominated by single-chain monomer fitness constraints, actively penalizing non-conserved interface residues that enhance binding to specific binding partners.

3.4 Evolutionary PPI Stratification: Homooligomer Self-Co-occurrence Fails

To test whether the lack of intermolecular co-evolution in single sequences explains the failure, we stratified the 2,262 interface variants into three distinct evolutionary regimes (Figure 5): 1. **Class 1:**

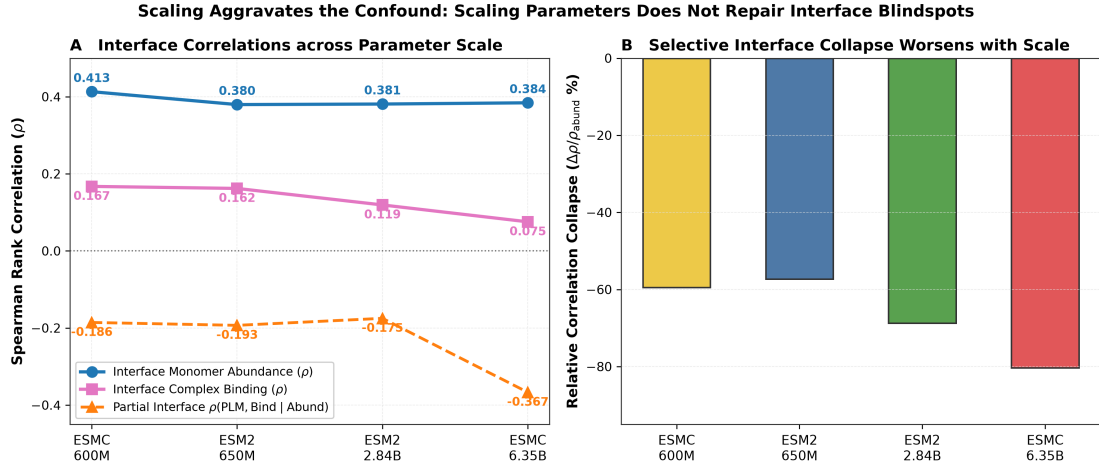


Figure 4: Model scaling exacerbates the quaternary interface defect. **(A)** As model scale increases from 600M to 6.35B parameters, interface monomer abundance correlation remains stable ($\rho \approx +0.38$ to $+0.41$), while complex binding correlation monotonically collapses from $+0.167$ down to $+0.075$, and partial correlation drops to -0.367 . **(B)** Relative correlation collapse ($\Delta\rho/\rho_{\text{abund}}$) deepens from -57.4% to -80.5% at 6.35B parameters. Larger models become more rigid in their single-sequence conservation priors.

Homooligomers (p53 / 10LG, $N = 513$ interface variants): Identical sequences in UniRef where intermolecular contact pairs co-occur within the exact same sequence. 2. **Class 2: Natural Co-evolving Heterodimers (HLA-A2 / 50PI, GB1 / 1FCC, $N = 1,011$ interface variants):** Intermolecular partners that have co-evolved over evolutionary timescales across species. 3. **Class 3: De Novo / Synthetic / Cross-Species (KRAS / 6H46 with DARPin K55, Spike RBD / 6M0J with human ACE2, $N = 738$ interface variants):** Non-natural or cross-species interfaces lacking mutual co-evolution.

Crucially, in **Homooligomers** (p53 homotetramer), where sequence self-co-occurrence is present in the training database, ESMC-6B displays severe **active anti-correlation** with binding affinity ($\rho = -0.565$, $\rho_{\text{partial}} = -0.511$, Figure 5). Single-chain attention mechanisms cannot distinguish whether a conserved contact residue is acting intra-molecularly or inter-molecularly, resulting in destructive interference between monomer tertiary folding and quaternary assembly energetics.

In **Natural Heterodimers** (HLA-A2, GB1), the apparent raw binding correlation ($\rho = +0.329$) is entirely mediated by monomer folding: controlling for abundance causes the partial correlation to drop near zero ($\rho_{\text{partial}} = +0.099$, and $+0.001$ on HLA-A2 alone). In **Synthetic Binders** (KRAS), the model exhibits no native prior for the synthetic DARPin partner, collapsing to $\rho_{\text{partial}} = -0.487$.

3.5 The “PLM Filter Trap” in Computational Binder Design

In computational protein engineering and therapeutic antibody design campaigns, zero-shot PLM likelihoods are standardly applied as an upstream **computational filter**: candidate mutations are scored, and only the top 10%–20% highest-likelihood variants are synthesized in the wet lab (Notin et al. 2023; Hie et al. 2022).

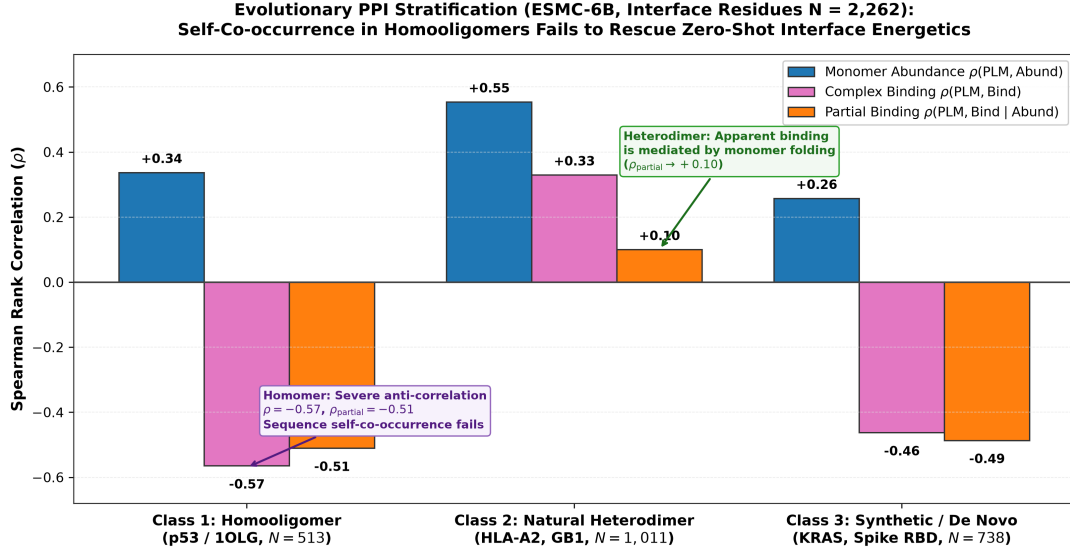


Figure 5: Evolutionary PPI stratification of interface variants ($N = 2,262$) evaluated on ESMC-6B. In homooligomers (p53), sequence self-co-occurrence fails to rescue interface energetics, resulting in severe anti-correlation ($\rho = -0.565$, $\rho_{\text{partial}} = -0.511$). In natural heterodimers (HLA-A2, GB1), controlling for expression reduces partial binding correlation to $\rho_{\text{partial}} = +0.100$. In synthetic binders (KRAS), single-chain PLMs are completely blind to non-natural partner physics ($\rho_{\text{partial}} = -0.487$).

We simulated this filtering workflow across all $N = 10,643$ variants ($N = 3,657$ experimentally beneficial binding variants, $N = 1,148$ beneficial interface variants, Table 4 and Figure 6).

Table 4: Quantitative simulation of the “PLM Filter Trap” in computational binder design across 4 foundation models ($N = 1,148$ experimentally validated beneficial interface mutations, $\Delta y_{\text{binding}} \geq 0.0$). Standard top-20% zero-shot likelihood filtering discards between 85.8% and 96.1% of true affinity-improving mutations.

Model Architecture	Filter Cutoff (Top X%)	Selected Interface Hits (N)	Interface False-Negative Rate (FNR)	Interface Depletion Rate vs. Non-Interface	Key Engineering Takeaway
ESM2-650M	Top 10%	112 / 1,148	90.2%	+2.9%	Discards 9 out of 10 true binding hits
	Top 20%	163 / 1,148	85.8%	+17.5%	Standard filter purges 85.8% of hits
ESM2-3B	Top 10%	105 / 1,148	90.8%	−8.8%	Severe depletion of interface mutations
	Top 20%	139 / 1,148	87.9%	+19.2%	Discards 87.9% of beneficial hits
ESMC-600M	Top 10%	72 / 1,148	93.7%	+2.9%	93.7% of true affinity hits purged

Model Architecture	Filter Cutoff (Top $X\%$)	Selected Interface Hits (N)	Interface False-Negative Rate (FNR)	Interface Depletion Rate vs. Non-Interface	Key Engineering Takeaway
ESMC-6B	Top 20%	99 / 1,148	91.4%	+32.2%	32.2% depletion relative to surface
	Top 10%	32 / 1,148	97.2%	+33.4%	Only 32 of 1,148 hits survive
	Top 20%	45 / 1,148	96.1%	+57.8%	Discards 96.1% of true interface hits

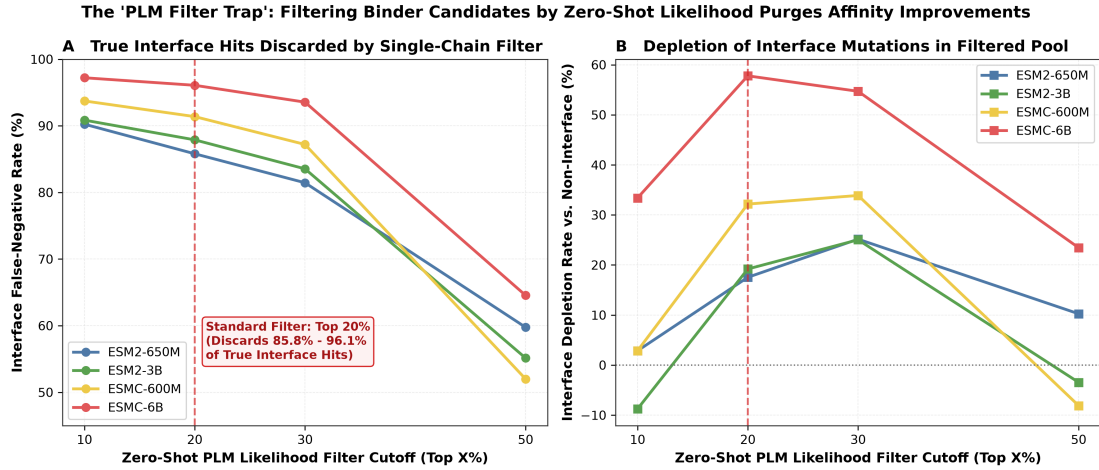


Figure 6: The “PLM Filter Trap” in computational protein and antibody engineering. **(A)** Interface False-Negative Rate (FNR) across zero-shot PLM filtering thresholds. Applying a standard top-20% likelihood filter discards 85.8% (ESM2-650M) to 96.1% (ESMC-6B) of experimentally validated affinity-improving interface mutations. **(B)** Interface Depletion Rate relative to non-interface mutations, showing that single-chain PLM filtering actively purges quaternary interface innovations from engineered libraries.

At standard engineering thresholds (selecting the top 20% by zero-shot score): - ESM2-650M discards 85.8% of true affinity-improving interface mutations (163/1,148 selected). - ESMC-6B discards **96.1%** of true affinity-improving interface mutations (45/1,148 selected). - The interface depletion rate reaches +57.8% relative to non-interface mutations, demonstrating that zero-shot PLMs actively purge affinity-enhancing mutations from candidate libraries.

3.5.1 1D Multi-Chain Concatenation Fails to Rescue Signal

We tested whether presenting the partner sequence in the context window via 1D concatenation (Target $\parallel$ $\langle \text{eos} \rangle \parallel$ Partner) restores interface sensitivity. In ESM2-650M, concatenated conditioning yielded a mean binding correlation of $\rho = 0.2202$ compared to $\rho = 0.2365$ for single-chain scoring (mean $\Delta\rho_{\text{binding}} =$

—0.0162). Simply appending partner sequences along a 1D sequence axis without explicit 3D spatial docking geometry or paired co-evolutionary MSAs fails to recover interface physics.

4 Discussion

The results presented here resolve an ongoing contradiction between structural biophysics and protein machine learning benchmarks. While single-chain protein language models have demonstrated transformative utility for predicting monomer folding, solubility, and thermal stability (Lin et al. 2023; Rives et al. 2021), their apparent success on protein-protein interaction benchmarks is an artifact of high-throughput display selection mechanics.

4.1 Architectural Limitations of 1D Sequence Transformers

Single-chain masked language models operate under two fundamental structural constraints: 1. **Absence of 3D Intermolecular Coordinates:** 1D self-attention matrices compute pairwise token interactions without explicit rigid-body relative transformation matrices ($\mathbf{R}, \mathbf{t}$) between interacting chains. 2. **Marginalization Over Monomer Evolution:** The training objective maximizes $\sum \log p(x_i | x_{\setminus i})$ across single UniRef entries. Evolution primarily conserves core tertiary packing and soluble surface charge patterns. When an interface residue mutates to improve affinity for an exogenous partner, the single-sequence prior treats this adaptation as an evolutionary anomaly, assigning it low likelihood.

4.2 Practical Recommendations for Computational Protein Design

Based on our findings, we formulate four concrete guidelines for protein and antibody engineering: 1. **Never use single-chain zero-shot PLM likelihoods as hard exclusionary filters** in binder design, antibody affinity maturation, or PPI optimization. Filtering at top 20% discards $> 90\%$ of true interface winners. 2. **De-conflate experimental DMS benchmarks:** Benchmark creators (e.g., ProteinGym) must report paired abundance-controlled metrics (ρ_{partial}) rather than raw binding correlations. 3. **Deploy structure-conditioned or complex-aware architectures:** Quaternary interface engineering requires explicit 3D complex models (e.g., AlphaFold-Multimer (Evans et al. 2022), ProteinMPNN (Dauparas et al. 2022), Rosetta (Alford et al. 2017), or FoldX (Guerois et al. 2002)) that evaluate intermolecular contact energetics in 3D coordinate space.

5 Methods

5.1 Within-Target Paired DMS Dataset Curation

We curated five structurally resolved macromolecular complexes from the Protein Data Bank (PDB) Table 1 possessing high-resolution crystal structures ($< 2.8 \text{ \AA}$) and paired single-mutant deep mutational scanning libraries in ProteinGym (Notin et al. 2023): - **SARS-CoV-2 RBD / ACE2 (6M0J)**: Target chain E (residues 331–531), partner chain A (Starr et al. 2020). - **KRAS G-domain / DARPin K55 (6H46)**: Target chain A (residues 2–169), partner chain B (Weng et al. 2024). - **HLA-A*02:01 / TAPBPR (50PI)**: Target chain A (residues 1–180), partner chains B, C (McShan et al. 2019). - **Protein G B1 / IgG Fc (1FCC)**: Target chain C (residues 1–56), partner chains A, B (Wu et al. 2016; Olson et al. 2014). - **p53 Tetramerization Domain (10LG)**: Target chain A (residues 325–356), partner chains B, C, D (Giacomelli et al. 2018).

Solvent-accessible surface area (SASA) was computed using FreeSASA v2.1.2 (Mitternacht 2016) with a $1.4 \text{ \AA}$ probe radius. Interface residues were defined by $\Delta\text{SASA} = \text{SASA}_{\text{monomer}} - \text{SASA}_{\text{complex}} > 0 \text{ \AA}^2$ or minimum heavy-atom distance $\leq 5.0 \text{ \AA}$ to partner chains (Table 5).

Table 5: Structural compartment definitions and variant counts across the paired DMS corpus.

Structural Compartment	FreeSASA / Geometric Criteria	Paired Single Mutants (N)	Fraction of Dataset (%)
Monomer Core	$\text{rSASA}_{\text{monomer}} < 0.20 \text{ \AA}^2 \wedge \Delta\text{SASA} = 0 \text{ \AA}^2$	4,405	41.4%
Monomer Surface	$\text{rSASA}_{\text{monomer}} \geq 0.20 \text{ \AA}^2 \wedge \Delta\text{SASA} = 0 \text{ \AA}^2$	3,976	37.4%
Quaternary Interface	$\Delta\text{SASA} > 0 \text{ \AA}^2 \vee d_{\text{partner}} \leq 5.0 \text{ \AA}$	2,262	21.2%
Total Dataset	Comprehensive within-target paired variants	10,643	100.0%

5.2 Zero-Shot PLM Likelihood Scoring

Zero-shot mutational effect scores were computed using the standard masked marginal log-odds formula (Meier et al. 2021):

$$\Delta \log p(x_{\text{mut}} | x) = \log p(x_i = \text{mut} | x_{\setminus i}) - \log p(x_i = \text{wt} | x_{\setminus i})$$

For ESM-2 models (esm2_t33_650M_UR50D and esm2_t36_3B_UR50D) (Lin et al. 2023), logits were extracted via PyTorch with float16 precision. For ESM-Cambrian models (esm2_600m and esm2_6b) (Hayes

et al. 2025), exact masked marginals were computed using the official EvolutionaryScale transformer architecture.

5.3 Three-Way Interaction Models and Clustered Sandwich Covariance

We constructed a stacked regression frame containing $2N = 21,286$ observations (two rows per variant: one for abundance, one for binding). Scores were standardized within each (system, assay) group:

$$z(y) = \frac{y - \bar{y}_{s,a}}{\sigma_{s,a}}$$

The three-way interaction model was estimated using Ordinary Least Squares:

$$\mathbf{y} = \mathbf{X}\beta + \epsilon$$

Cluster-robust standard errors were computed using the Liang-Zeger sandwich estimator clustered across the $G = 5$ macromolecular systems (Liang and Zeger 1986):

$$\mathbf{V}_{\text{cluster}} = \frac{G}{G-1} \frac{N-1}{N-K} (\mathbf{X}^T \mathbf{X})^{-1} \left(\sum_{g=1}^G \mathbf{X}_g^T \hat{\epsilon}_g \hat{\epsilon}_g^T \mathbf{X}_g \right) (\mathbf{X}^T \mathbf{X})^{-1}$$

5.4 Non-Parametric Permutation and Small-Cluster Econometric Inference

Because asymptotic cluster-sandwich estimators with $G = 5$ clusters have only $df = 4$, we employed within-system stratified permutation tests ($B = 10,000$ iterations) as our primary exact non-parametric test. In each permutation b , the structural interface indicator was shuffled within each system, preserving system-level variance and marginal distributions while breaking interface-specific association:

$$p_{\text{perm}} = \frac{1}{B} \sum_{b=1}^B \mathbb{I}(|\hat{\beta}_7^{(b)}| \geq |\hat{\beta}_7^{\text{obs}}|)$$

In addition, we computed small- G Wild Cluster Bootstrap p -values using the Webb 6-point distribution ($w_g \in \{-\sqrt{1.5}, -1.0, -\sqrt{0.5}, \sqrt{0.5}, 1.0, \sqrt{1.5}\}$) (Webb 2023; Cameron et al. 2008) with $B = 10,000$ draws, and fit system fixed-effects OLS models ($z(\text{DMS}) \sim \text{PLM} \times \text{Binding} \times \text{Interface} + \sum_s \gamma_s \mathbb{I}_{\text{System}=s}$).

5.5 Structural Definition Sensitivity Sweep Grid

We evaluated structural definition robustness across a 5×5 parameter grid:

$$\Delta\text{SASA}_{\text{cutoff}} \in \{2.0, 5.0, 10.0, 15.0, 20.0\} \text{ \AA}^2, \quad d_{\text{min, cutoff}} \in \{3.5, 4.0, 4.5, 5.0, 6.0\} \text{ \AA}$$

For each grid cell, residues satisfying $\Delta\text{SASA} \geq \Delta\text{SASA}_{\text{cutoff}} \vee d_{\text{min}} \leq d_{\text{min, cutoff}}$ were labeled interface, and non-interface residues were split into Core ($\text{rSASA} < 0.20$) and Surface ($\text{rSASA} \geq 0.20$).

5.6 Hierarchical Evolutionary Class Modeling with HC3 Robust Covariance

In the cross-class hierarchical interaction model ($z(\text{DMS}) \sim \text{PLM} \times \text{Binding} \times \text{Class}$ at interface positions), evolutionary class is constant within each system cluster. To prevent collinear cluster-score covariance, we computed MacKinnon & White HC3 heteroskedasticity-consistent standard errors (MacKinnon and White 1985):

$$\mathbf{V}_{\text{HC3}} = (\mathbf{X}^T \mathbf{X})^{-1} \left(\sum_{i=1}^N \frac{\hat{e}_i^2}{(1 - h_{ii})^2} \mathbf{x}_i \mathbf{x}_i^T \right) (\mathbf{X}^T \mathbf{X})^{-1}$$

where $h_{ii} = \mathbf{x}_i^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}_i$ is the i -th diagonal leverage element.

5.7 Causal Mediation and Partial Rank Correlation

To evaluate mediation, we computed the partial Spearman rank correlation by residualizing rank-transformed variables:

$$\mathbf{r}_{\text{PLM}} = \text{rank}(\text{PLM}), \quad \mathbf{r}_{\text{Bind}} = \text{rank}(\text{Binding}), \quad \mathbf{r}_{\text{Abund}} = \text{rank}(\text{Abundance})$$

Residuals $\mathbf{e}_{\text{PLM}} = \mathbf{r}_{\text{PLM}} - \hat{\mathbf{r}}_{\text{PLM}}(\mathbf{r}_{\text{Abund}})$ and $\mathbf{e}_{\text{Bind}} = \mathbf{r}_{\text{Bind}} - \hat{\mathbf{r}}_{\text{Bind}}(\mathbf{r}_{\text{Abund}})$ were obtained via univariate linear regression. The partial correlation is the Pearson correlation between these residuals:

$$\rho(\text{PLM}, \text{Binding} \mid \text{Abundance}) = \frac{\mathbf{e}_{\text{PLM}}^T \mathbf{e}_{\text{Bind}}}{\|\mathbf{e}_{\text{PLM}}\| \|\mathbf{e}_{\text{Bind}}\|}$$

5.8 Multi-Chain Concatenated Conditioning

In multi-chain evaluation, target and partner sequences were concatenated with an end-of-sequence delimiter:

$$\mathbf{s}_{\text{complex}} = \mathbf{s}_{\text{target}} \parallel \langle \text{eos} \rangle \parallel \mathbf{s}_{\text{partner}}$$

Masked marginal log-odds were evaluated at target interface positions while conditioning on the full concatenated sequence context in ESM2-650M.

6 Data and Code Availability

<https://github.com/emiliovenegas/plm-quaternary-interfaces>

References

- Alford, Rebecca F., Andrew Leaver-Fay, Jeliasko R. Jeliaskov, et al. 2017. “The Rosetta All-Atom Energy Function for Macromolecular Modeling and Design.” *Journal of Chemical Theory and Computation* 13 (6): 3031–48. <https://doi.org/10.1021/acs.jctc.7b00125>.
- Cameron, A. Colin, Jonah B. Gelbach, and Douglas L. Miller. 2008. “Bootstrap-Based Improvements for Inference with Clustered Errors.” *The Review of Economics and Statistics* 90 (3): 414–27. <https://doi.org/10.1162/rest.90.3.414>.
- Dauparas, Justas, Ivan Anishchenko, Nathaniel Bennett, et al. 2022. “Robust Deep Learning-Based Protein Sequence Design Using ProteinMPNN.” *Science* 378 (6615): 49–56. <https://doi.org/10.1126/science.add2187>.
- Evans, Richard, Michael O’Neill, Alexander Pritzel, et al. 2022. “Protein Complex Prediction with AlphaFold-Multimer.” *bioRxiv*, ahead of print. <https://doi.org/10.1101/2021.10.04.463034>.
- Giacomelli, Andrew O., Xiaoping Yang, Robert E. Lintner, et al. 2018. “Mutational Processes Shape the Landscape of TP53 Mutations in Human Cancer.” *Nature Genetics* 50 (10): 1381–87. <https://doi.org/10.1038/s41588-018-0204-y>.
- Guerois, Raphaël, Jens Erik Nielsen, and Luis Serrano. 2002. “Predicting Changes in the Stability of Proteins and Protein Complexes: A Study of More Than 1000 Mutations.” *Journal of Molecular Biology* 320 (2): 369–87. [https://doi.org/10.1016/S0022-2836\(02\)00442-4](https://doi.org/10.1016/S0022-2836(02)00442-4).
- Hayes, Thomas, Roshan Rao, Halil Akin, et al. 2025. “Simulating 500 Million Years of Evolution with a Language Model.” *Science* 387 (6736): 850–58. <https://doi.org/10.1126/science.ads0018>.
- Hie, Brian L., Kevin K. Yang, and Peter S. Kim. 2022. “Evolutionary Velocity with Protein Language Models Predicts Evolutionary Dynamics of Diverse Proteins.” *Cell Systems* 13 (4): 274–285.e6. <https://doi.org/10.1016/j.cels.2022.01.003>.
- Jankauskaitė, Justina, Brian Jiménez-García, Justas Dapkūnas, Juan Fernández-Recio, and Iain H. Moal. 2019. “SKEMPI 2.0: An Updated Benchmark of Changes in Protein–Protein Binding Energy, Kinetics and Thermodynamics Upon Mutation.” *Bioinformatics* 35 (3): 462–69. <https://doi.org/10.1093/bioinformatics/bty635>.
- Liang, Kung-Yee, and Scott L. Zeger. 1986. “Longitudinal Data Analysis Using Generalized Linear Models.” *Biometrika* 73 (1): 13–22. <https://doi.org/10.1093/biomet/73.1.13>.
- Lin, Zeming, Halil Akin, Roshan Rao, et al. 2023. “Evolutionary-Scale Prediction of Atomic-Level

- Protein Structure with a Language Model.” *Science* 379 (6637): 1123–30. <https://doi.org/10.1126/science.ade2574>.
- MacKinnon, James G., and Halbert White. 1985. “Some Heteroskedasticity-Consistent Covariance Matrix Estimators with Improved Finite Sample Properties.” *Journal of Econometrics* 29 (3): 305–25. [https://doi.org/10.1016/0304-4076\(85\)90158-7](https://doi.org/10.1016/0304-4076(85)90158-7).
- McShan, Andrew C., Christine A. Devlin, Sarah A. Overall, et al. 2019. “Molecular Determinants of Chaperone Interactions on MHC-I for Folding and Antigen Repertoire Selection.” *Proceedings of the National Academy of Sciences* 116 (51): 25602–13. <https://doi.org/10.1073/pnas.1915562116>.
- Meier, Joshua, Roshan Rao, Robert Verkuil, Jason Liu, Tom Sercu, and Alexander Rives. 2021. “Language Models Enable Zero-Shot Prediction of the Effects of Mutations on Protein Function.” In *Advances in Neural Information Processing Systems*, edited by M. Ranzato, A. Beygelzimer, Y. Dauphin, P. S. Liang, and J. Wortman Vaughan, vol. 34. Curran Associates, Inc. https://proceedings.neurips.cc/paper_files/paper/2021/hash/f51338d736f95dd42427296047067694-Abstract.html.
- Mitternacht, Simon. 2016. “FreeSASA: An Open Source C Library for Solvent Accessible Surface Area Calculations.” *F1000Research* 5: 189. <https://doi.org/10.12688/f1000research.7931.1>.
- Notin, Pascal, Aaron Kollasch, Daniel Ritter, et al. 2023. “ProteinGym: Large-Scale Benchmarks for Protein Fitness Prediction and Design.” *Advances in Neural Information Processing Systems* 36: 64331–79. https://proceedings.neurips.cc/paper_files/paper/2023/file/cac723e5ff29f65e3fcbb0739ae91bee-Paper-Datasets_and_Benchmarks.pdf.
- Olson, C. Anders, Nicholas C. Wu, and Ren Sun. 2014. “A Comprehensive Biophysical Description of Pairwise Epistasis Throughout an Entire Protein Domain.” *Current Biology* 24 (22): 2643–51. <https://doi.org/10.1016/j.cub.2014.09.072>.
- Rives, Alexander, Joshua Meier, Tom Sercu, et al. 2021. “Biological Structure and Function Emerge from Scaling Unsupervised Learning to 250 Million Protein Sequences.” *Proceedings of the National Academy of Sciences* 118 (15): e2016239118. <https://doi.org/10.1073/pnas.2016239118>.
- Sledzieski, Samuel, Rohit Singh, Lenore Cowen, and Bonnie Berger. 2021. “D-SCRIPT Translates Genome to Phenome with Sequence-Based, Structure-Aware, Genome-Scale Predictions of Protein-Protein Interactions.” *Cell Systems* 12 (10): 969–982.e6. <https://doi.org/10.1016/j.cels.2021.08.010>.
- Starr, Tyler N., Allison J. Greaney, Sarah K. Hilton, et al. 2020. “Deep Mutational Scanning of SARS-CoV-2 Receptor Binding Domain Reveals Constraints on Folding and ACE2 Binding.” *Cell* 182 (5): 1295–1310.e20. <https://doi.org/10.1016/j.cell.2020.08.012>.

- Webb, Matthew D. 2023. “Reworking Wild Bootstrap-Based Inference for Clustered Errors.” *Canadian Journal of Economics / Revue Canadienne d’économie* 56 (3): 839–58. <https://doi.org/10.1111/caje.12661>.
- Weng, Chenchun, Andre J. Faure, Albert Escobedo, and Ben Lehner. 2024. “The Energetic and Allosteric Landscape for KRAS Inhibition.” *Nature* 626 (7999): 643–52. <https://doi.org/10.1038/s41586-023-06954-0>.
- Wu, Nicholas C., Lei Dai, C. Anders Olson, James O. Lloyd-Smith, and Ren Sun. 2016. “Adaptation in Protein Fitness Landscapes Is Facilitated by Indirect Paths.” *eLife* 5: e16965. <https://doi.org/10.7554/eLife.16965>.